Fiscal Year:	FY 2005	**Task Last Updated:**	FY 08/02/2005
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Project Title:	Advanced Optical Frequency Standard for Space		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94305-4008	**Congressional District:**	15
Comments:	NOTE: Previously at National Institute of Standards & Technology (Ed., Fall 2011)		
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	05/01/2001	**End Date:**	01/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	C73540A		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		

To develop high-accuracy high-stability optical frequency standards for space based on laser-cooled and trapped neutral atoms and ions. Microwave frequency standards have achieved higher stability and accuracy in the last decade due in large part to laser cooling and trapping [1,2,3]. Better stability and accuracy will be difficult since microwave clocks are now near their fundamental performance limit. However, if the frequency of the standard could be made higher, then its stability (and possibly its accuracy) could be proportionally higher [4]. Thus, optical oscillators that are referenced to narrow resonances in laser-cooled atoms will be the time and frequency standards of the future. Our research focuses on eliminating or reducing the most serious impediments to the performance of optical frequency standards.
These include: 1.Acceleration-induced noise in optical reference cavities 2.Gravitation limitation to the interaction time for neutral-atom clocks 3.An optical frequency synthesis/measurement system that preserves the phase fidelity of the optical references

Task Description:	We are evaluating the performance of two optical frequency standards, one based on a trapped and laser-cooled single Hg+ ion and the other on laser-cooled calcium atoms. Since the fractional frequency instability of atomic clocks is inversely proportional to the frequency of the clock transition and to the square root of the Ramsey interrogation period, greatly improved performance compared to microwave frequency standards is possible by utilizing a narrow optical transition in laser-cooled atoms. Projected fractional frequency instabilities and inaccuracies for laser-cooled, stored ion optical frequency standards that are earth based are about 1x10-18, more than 2 orders of magnitude better than that projected for atomic microwave frequency standards even in a micro-gravity environment. However, optical frequency standards based on laser-cooled neutral atoms in a micro-gravity environment could potentially produce even higher stabilities. In order to realize their projected stabilities, both types of optical frequency standards require a spectrally pure local oscillator that has a short term (<10 s) instability less than 5x10-16. Most often, this is achieved by locking the frequency of the laser to a high-finesse and well-isolated optical cavity. Frequency fluctuations of the optical oscillator are then dominantly caused by seismic perturbations that are coupled through the suspension system which must support the cavity against the pull of gravity. Since the fundamental resonance of the isolation support system is proportional to the sqrt(g, seismic isolation could be more than 3 orders of magnitude better in a micro-g environment. Finally, the full benefits of a highly stable and accurate optical frequency standard are available only by counting the zero crossings of the frequency of the optical standard to generate time. We plan to thoroughly investigate broadband optical synthesis schemes using pulsed lasers to connect visible frequencies to countable microwave frequencies in a phase coherent manner. Recent progress in our laboratory and others shows that mode-locked fs-lasers coupled to photonic fibers can be used to span large frequency intervals in the visible spectrum and to measure optical frequencies with high precision and accuracy. The many orders of magnitude improvement expected in the performance of frequency standards based on optical transitions will open many new opportunities in experimental fundamental physics, including: improved tests of relativity, searches for time variation of fundamental constants, precision time and frequency metrology. We can also anticipate applications in advanced communication systems and navigation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We have made good progress in FY05 on our NASA funded project for optical frequency standards for space. For the mercury optical clock we have made two more absolute frequency measurements in direct comparison to the Cs fountain standard NIST-F1. The combined statistical and systematical uncertainties were about 1.5 × 10^-15 in January ’05 and about 1 × 10^-15 in June ’05. The total systematic uncertainty for the mercury standard in the most recent measurement in June was conservatively estimated to be < 1 × 10^-16, a factor of 2 smaller than the systematic uncertainty of NIST-F1. The reduction in the uncertainty of the mercury standard resulted primarily from an absolute measurement of the electric quadrupole moment. Furthermore, by applying a magnetic field of constant magnitude successively in three orthogonal directions, we effectively eliminate the quadrupole shift and do not expect it to limit the accuracy at the 10^-18 level. Our two recently completed measurements reveal no variation greater than 1 part in 10^15 over the past six months. Both limits constrain the temporal variation of the fine structure constant alpha to less than 3 x 10^-16/yr. To date, this is the most stringent laboratory test of the time stability of alpha. Additionally in the past year there has been considerable development of the two-ion, “quantum logic” optical clock. In this system, two ions of different species are held in a linear RF trap. In the version being pursued at NIST, we are using an aluminum clock ion and a beryllium control ion, and we have begun to evaluate the performance and suitability of this system as a next-generation optical frequency standard. During the last year, we have optimized the cooling process and coherent state manipulation of the beryllium ion and demonstrated the basic protocol on the more allowed 1S0 - 3P1 transition in Al+ as well as the narrow 1S0 - 3P0 optical clock transition. Initial measurements vs. the mercury standard show that reproducibility of the ratio of the frequencies is ~ 1 part in 10^16 over a month-long period and we expect that this will be another order of magnitude better very soon. For the optical clock based on laser-cooled neutral Ca atoms, with the dominant systematic shifts due to Doppler effects now virtually removed from the apparatus, we have performed an absolute frequency measurement of the clock transition at 657 nm. The final fractional uncertainty was 7.5x10^-15, the most accurate measurement of a neutral atom frequency to date. Presently we are evaluating the achievable stability with a greatly simplified version of the Ca clock with an eye on applications. We have also made considerable progress on a second optical clock, based on laser-cooled neutral Yb, which has enormous potential for high accuracy and stability. We have achieved first and second-stage cooling for both the relevant isotopes with resultant atomic cloud temperatures of 10’s of microkelvin. We will soon be loading the atoms into a specially designed optical lattice, which will greatly reduce systematic shifts for the clock transition at 578 nm, which has a natural linewidth of only 10 mHz. To expedite spectroscopy in the lattice, we performed the first absolute frequency measurements of the clock transitions, by using atoms released from the second-stage magneto-optic trap. The frequency uncertainty for these transitions was reduced nearly one-million-fold to 4.4 kHz, limited by Doppler issues that will be absent in the lattice environment.
Bibliography Type:	Description: (Last Updated: 09/03/2025)
Articles in Peer-reviewed Journals	Oskay WH, Itano WM, Bergquist JC. "Measurement of the 199Hg+5d(9)6s(2) (2)D5/2 electric quadrupole moment and a constraint on the quadrupole shift." Physical Review Letters. 2005 Apr;94(16):163001. http://dx.doi.org/10.1103/PhysRevLett.94.163001 , Apr-2005
Articles in Peer-reviewed Journals	Oates CW, Wilpers G, Hollberg L. "Observation of large atomic-recoil-induced asymmetries in cold atom spectroscopy." Physical Review A. 2005 Feb;71(2):023404. http://dx.doi.org/10.1103/PhysRevA.71.023404 , Feb-2005
Articles in Peer-reviewed Journals	Sterr U, Degenhardt C, Stoehr H, Lisdat C, Schnatz H, Helmcke J, Riehle F, Wilpers G, Oates C, Hollberg L. "The optical calcium frequency standards of PTB and NIST." Comptes Rendus Physique. 2004 Oct;5(8):845-55. http://dx.doi.org/10.1016/j.crhy.2004.08.005 , Oct-2004
Articles in Peer-reviewed Journals	Hoyt CW, Barber ZW, Oates CW, Fortier TM, Diddams SA, Hollberg L. "Observation and absolute frequency measurements of the 1S0-3P0 optical clock transition in neutral ytterbium." Physical Review Letters. 2005 Aug;95(8):083003. http://dx.doi.org/10.1103/PhysRevLett.95.083003 , Aug-2005